Capital University of Science and Technology

_____ _____Department of_Software Engineering

MTAI2053-Multi-Variate Calculus

ASSIGNMENT NO.1

Semester: Spring 2026 Max Marks: 10

Instructor: Dr.Ayyesha Kanwal Ahmed Due Date: March 25, 2026

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Q1:-

Question No.1:Show that $\lim_{y)\to(0,0)} \frac{\sqrt{x^2y^2+1}-1}{x^2+y^2} = 0.$

Sol:-

$$\frac{\sqrt{(l\cos\theta)^2(l\sin\theta)^2+1}-1}{(l\cos\theta)^2+(l\sin\theta)^2}$$

$$= \frac{\sqrt{l^4-\sin^2\theta\cos^2\theta+1}-1}{l^2(\cos^2\theta+\sin^2\theta)}$$

$$= \frac{\sqrt{l^4\cos^2\theta\sin^2\theta+1}-1}{l^2} \times \frac{\sqrt{l^4\cos^2\theta\sin^2\theta+1}-1}{\sqrt{l^4\cos^2\theta\sin^2\theta+1}+1}$$

$$= \sqrt{\frac{l^4\cos^2\theta\sin^2\theta-(1)^2}{l^2(l^4\text{pos}^2\theta\sin^2\theta+1+1)}}$$

$$= \frac{l^2}{l^2}\frac{\cos^2\theta\sin^2\theta+X}{(\sqrt{l^4\text{pos}^2\theta\sin^2\theta}+1+1)}$$

$$= \frac{l^2\cos^2\theta\sin^2\theta}{\sqrt{l^4\cos^2\theta\sin^2\theta}+1+1}$$

$$\lim_{(x,y)\to(0,0)} \frac{(0^2)\cos^2(0)\cdot\sin^2\theta}{\sqrt{(0)^4\cdot\cos^2(0)\sin^2(0)+1+1}}$$

$$= \frac{0\cdot 1\cdot 0}{\sqrt{0\cdot 1\cdot 0+1+1}}$$

$$= \frac{0}{\sqrt{1+1}}$$

$$= \frac{0}{2}$$

$= 0$ Ans

Q2:-

i) Along y-axis $x=0$;

$$\Rightarrow \lim_{y\to 0} \frac{0+y^3}{0-y}$$

$$\Rightarrow \lim_{y\to 0} \frac{y^3}{y}$$

$$\Rightarrow \lim_{y\to 0} (-y^2)$$

$$\Rightarrow 0$$

ii) Along x-axis $y=0$;

$$\Rightarrow \lim_{x\to 0} \frac{x^3+(0)^2}{x-0}$$

$$\Rightarrow \lim_{x\to 0} \frac{x^3}{x}$$

$$\Rightarrow \lim_{x\to 0} x^2$$

$$= (0)^2 = 0$$

iii) Along $y = x - x^3$

$$\lim_{x \to 0} = \frac{x^3 + (x - x^3)^3}{x - (x - x^3)}$$

$$\lim_{x \to 0} \frac{x^3 + (x - x^3)^3}{\cancel{x} - \cancel{x} + x^3}$$

$$\lim_{x \to 0} \frac{x^3 + (x - x^3)^3}{x^3}$$

$$\lim_{x \to 0} \frac{x^3}{x^3} + \frac{(x - x^3)^3}{x^3}$$

$$\lim_{x \to 0} \left(1 + \left(\frac{x - x^3}{x}\right)^3\right)$$

$$\lim_{x \to 0} \left(1 + (1 + x^{3-1})^3\right)$$

$$\lim_{x \to 0} \left(1 + (1 - x^2)^3\right)$$

$$= (1 + (1 - 0)^3)$$

$$= 1 + 1$$

$$= 2$$

$$2 \neq 0$$

limit doesn't exits.